Title

Basic Computation Practice.

Background

N/A

Program Specifications

Implement a program with the following functions:

- Output the first primes of the inputted number.
- Check a Fibonacci number.
- Sum of digits of a positive natural number.

Function details:

1. Function 1: Display program instruction and prompt users to select an option from 1 to 3.
   - Users run the program. The program displays instruction and asks users to select an option from 1 to 3:
     - 1 – output first primes of the inputted number
     - 2 – check a Fibonacci number
     - 3 – sum of digits in a positive natural number
   - If user inputs 1, perform Function 2.
   - If user inputs 2, perform Function 3.
   - If user inputs 3, perform Function 4.
2. Function 2: Output first primes of the inputted number
   - Prompt users to input a number (n) which is 0 < n <= 50. Users re-input if n is not in valid range.
   - Output primes from 1 to n.
   - Go back to Function 1.
3. Function 3: Check a Fibonacci number.
   - Ask users to input a number (n) which is 0 < n <= 1000. Users re-input if n is not in valid range.
   - Return 1 if the inputted number is a Fibonacci number. Otherwise, return 0
   - Go back to Function 1.
4. Function 4: Sum of digits in a positive natural number.
   - Ask users to input a positive natural number. Users re-input the number if it is not valid.
   - Output sum of the digits in the inputted number.
   - Go back to Function 1.

Expectation of User interface:

```
1-The first primes
2-Fibonacci element
3-Sum of digits
Choose an option:1
Number of primes:5
2 3 5 7 11
1-The first primes
2-Fibonacci element
3-Sum of digits
Choose an option:
```

Output first primes of the inputted number 'n'

```
1-The first primes
2-Fibonacci element
3-Sum of digits
Choose an option:2
Number tested:21
It's a Fibonacci term.

1-The first primes
2-Fibonacci element
3-Sum of digits
Choose an option:2
Number tested:20
It's not a Fibonacci term.
```

Check a Fibonacci number.

```
1-The first primes
2-Fibonacci element
3-Sum of digits
Choose an option:3
Enter an integer:123
Sum of it's digits:6
```

Sum of digits in a natural number

Guidelines

Definition:

The Fibonacci Sequence is the series of numbers: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ... The next number is found by adding up the two numbers before it.

For example, 2 is found by adding the two numbers before it (1+1). Similarly, 3 is found by adding the two numbers before it (1+2)

Regression formula of Fibonacci sequence

$$F(n) := \begin{cases} 1, \text{ when } n = 1 \\ 1, \text{ when } n = 2 \\ F(n-1) + F(n-2) \text{ when } n > 2 \end{cases}$$

Example: 1, 1, 2, 3, 5, 8, 13, 21 … are first Fibonacci numbers.

A prim is a natural number divisible only by 1 and itself. 0 and 1 is not considered a prime number

Primes from 2 to 100:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97